Introduction to Calculus for Physics

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1 Introduction

This primer is meant to serve as an introduction to the calculus necessary to be successful in physics. No prior knowledge of calculus is assumed, however mastery of elementary algebra and trigonometry will be necessary for your success in this course. This text, supplemented with an AP Calculus course will be more than enough to prepare for the required mathematics of AP Physics.

A Note to the Reader

AP Physics is an exciting but demanding course. The level of mathematical and scientific maturity and sophistication required for success in this course will most likely be beyond anything you've encountered in your academic career so far. That said, for those of you who put in the hard work to master the material in this course, the rewards will be bountiful. The immediate reward will be the potential to earn college credit for up to two calculus-based physics courses. Not only will this give you a head start in college over most of your peers, it will also give you the potential to save a lot of money. College credit being granted is dependent on the university you choose to attend, but most universities will grant you credit if you score a 4 or a 5 on the AP exam. To give you a feel for how much money you could potentially save, I'll use my alma-mater, The University of Washington as an example: The tuition cost of 10 credits for an out of state student attending the University of Washington during the 2014-2015 academic year is $\$11,172.00$, so earning a 4 or 5 on both the AP Physics C: Mechanics and AP Physics C: Electromagnetism exams would not only allow you to skip two college physics classes, it would also save you over $\$11,000$! Remember, this is just the short term reward. In the long term, mastery of the concepts covered in this class can provide you with a deeper understanding of how the world works around you and serve as an entry point to careers ranging from engineering, medicine, and research science, to business and finance, and even to law and government. The applications are virtually endless!

How to Read this Primer

This primer is written in a similar format to how many college-level texts in pure mathematics are written (this is to help prep you for the rigors of reading a college-level math or science text). Each definition, theorem, and example given is properly marked for organizational purposes. As you read through the sections of this text, I highly suggest that you write down each definition given to you, and pay close attention to the wording to make sure you understand the definition. Sometimes it helps to draw a picture or a diagram of the definition so you can visualize it better. 30 example problems are embedded in this text and I would *strongly* suggest that for each example you encounter, don't just simply read the example, but try it out for yourself and see how far you can get. If you get stuck, look closely at the steps outlined in the text, follow them, and do your best to make sure you understand them. Eventually you'll get to

the solution. It is okay if you aren't able to answer an example problem on your own, but giving them an attempt and closely following each step of the solution (making sure that you understand each step, too), will do wonders for your problem solving skills and conceptual understanding of the content given to you. Now that all of this has been laid out for you, get ready to do science like you've never done it before!

2 Limits

2.1 Definition and Examples

The notion of a limit is one of the most important, and arguably most difficult concepts in all of calculus. Fortunately for us, in order to use limits in physics, we do not need to come up with a rigorous formulation of the concept of a limit, which makes our work much easier. For the purposes of this class, we will define a limit as follows:

Definition 1. *Suppose a function $f(x)$ is defined when x is close to the number a. Then we write*

$$\lim_{x \to a} f(x) = L$$

and say "the limit of $f(x)$, as x approaches a, equals L."

In general, evaluating a limit, is no different than evaluating a function (with a few minor exceptions, one of which will be shown in example 2).

Example 1. *Find the value of*

$$\lim_{x \to 4} x^3 + 4x^2 - 11$$

Solution: Since x is approaching 4, we can simply plug in 4 for x to evaluate this limit, thus

$$\lim_{x \to 4} x^3 + 4x^2 - 11 = \lim_{x \to 4} (4)^3 + 4(4)^3 - 11 = 117,$$

so in this case (as is often the case), the limit is just equal to $f(4)$.

□

Sometimes limits are defined where a function is not defined, because we are looking for the value that $f(x)$ *approaches* as x *approaches* a certain value. This subtlety won't be of much importance in physics, but the following example will help to illustrate this point.

Example 2. *Find the value of*

$$\lim_{x \to 1} \frac{x - 1}{x^2 - 1}$$

Solution: Some of you may recall that this function has a removable discontinuity (a hole) at $x = 1$. Normally, when evaluating limits, you can simply plug the value that x approaches into your function. Unfortunately, in this case, if you plug 1 in for x, you will be left with

$$\frac{x-1}{x^2-1} = \frac{(1)-1}{(1)^2-1} = \frac{0}{0}$$

which is undefined. One way you can evaluate this limit is by looking at values where x is very close to 1. For instance, if you set x to be 0.9999, you will get

$$\frac{x-1}{x^2-1} = \frac{(0.9999)-1}{(0.9999)^2-1} = 0.500025$$

and if you set x to be 1.0001, you will get

$$\frac{x-1}{x^2-1} = \frac{(1.0001)-1}{(1.0001)^2-1} = 0.499975.$$

Clearly, one could infer at this point that

$$\lim_{x\to 1} \frac{x-1}{x^2-1} = 0.5,$$

which is the correct answer.

□

What we just showed here is that evaluating the limit as a function approaches a particular value is not always the same as evaluating the function at the same value. In our example, $f(1)$ was undefined because there was a removable discontinuty, but the *limit* as x approaches 1 of $f(x)$ does exist and is equal to 0.5.

2.2 Exercises

Calculate each limit. For cases where the answer is undefined when you substitute the value where the limit is being evaluated into the equation, plug in values close to the value where the limit is being evaluated and you should get a well defined answer.

1. $$\lim_{x\to 2} x - 2$$

2. $$\lim_{k\to -6} \frac{k+6}{k-6}$$

3. $$\lim_{r\to 0} \frac{7r^2 - 3r + 1}{8r^3 + 4r^2 - 2}$$

4.
$$\lim_{x\to 0} \frac{3x^2}{x}$$
(*hint*: simplify this equation algebraically first).

5.
$$\lim_{t\to 0} \frac{\sqrt{t^2+16}-4}{t^2}$$

6.
$$\lim_{x\to 0} \cos\left(\frac{4}{\pi}x\right)$$

7.
$$\lim_{x\to 0} \frac{\sin(x)}{x}$$

8.
$$\lim_{y\to 9} \sqrt{\frac{y-8}{y^2-8y-7}}$$

9.
$$\lim_{\alpha\to 4} \frac{2\alpha^2-6\alpha-8}{3\alpha^2-9\alpha-12}$$

2.3 Infinite Limits

Sometimes limits of functions can become arbitrarily large or arbitarily small, to the point that the limit is infinite. Example 3 below shows just this.

Example 3. *Evaluate*

$$\lim_{x \to 0} \frac{1}{x^2}$$

Solution: The table below shows some values of this limit as we plug in values of x close to 0

x	$1/x^2$
1	1
0.1	100
0.01	10000
0.001	1000000

Table 1: Some values of $1/x^2$ as x tends toward 0. Notice that $1/x^2$ asymptotically increases as x goes to 0.

As you can see, when we make x closer and closer to 0, $1/x^2$ gets bigger and bigger. In fact, it turns out that this limit diverges to infinity, so we write

$$\lim_{x \to 0} 1/x^2 = \infty.$$

□

It is important to recognize that ∞ itself is not a real number. This is why you have always been taught that $\frac{1}{0}$, for instance, is undefined, and not equal to infinity. With the case of limits however, when we say that the limit as x approaches 0 of $1/x^2$ is equal to ∞, we are simply saying that as we take x to 0, our function, $f(x)$, increases without bound toward infinity, so in other words, $f(x)$ approaches but does not hit infinity. We are now equipped to introduce another definition:

Definition 2. *Let f be a function defined on both sides of a, except possibly at a itself. Then*

$$\lim_{x \to a} f(x) = \infty$$

means that the values of $f(x)$ can be made as large as we please by taking x sufficiently close to a, but not equal to a.

Conversely, we can also take limits "at" infinity. Example 4 illustrates this:

Example 4. *Compute*

$$\lim_{x\to\infty} \frac{1}{x}.$$

Solution: In this example, we are seeing what happens to the function $f(x) = 1/x$ as x tends to infinity. This can be computed by plugging in values for x. The table below shows some of these values:

x	$1/x$
1	1
100	0.01
10000	0.0001
1000000	0.000001

Table 2: Some values of $1/x$ as x tends to infinity. Notice that as x increases, $1/x$ asymptotically decreases toward 0.

Looking at the values in this table, it is reasonable to assume that $1/x \to 0$ as $x \to \infty$, so it follows that

$$\lim_{x\to\infty} 1/x = 0.$$

□

At this point, we should have enough familiarity for what a limit is to begin to dive into some remarkably useful concepts in calculus. You will encounter limits into a much greater amount of depth in AP Calculus, so you will have that to look forward to.

3 The Derivative

3.1 Introductary Remarks

The **derivative** forms the basis of the branch of mathematics called **differential calculus**. Differential calculus is concerned with rates of change and is an indispensible tool for physics. Last year, you found that many quantities in physics are defined to be the rate of change of something with respect to time. Examples include: velocity (rate of change of displacement with respect to time), acceleration (rate of change of velocity over time), force (rate of change of momentum over time), power (rate of change of work over time), current (rate of change of charge over time), and so on. It turns out each of these quantities can be defined by a derivative, so without further ado, we will define the derivative.

Definition 3. *The derivative of a function f at value x, is denoted by $f'(x)$, and is given by*

$$f'(x) = \lim_{h\to 0} \frac{f(x+h) - f(x)}{h}.$$

3.2 Computing Derivatives and More Notation

Before we dive deeper into the theory behind the derivative, let's look at some examples.

Example 5. *Compute the derivative of $f(x) = x^2$ with respect to the variable x.*

Solution: The solution to this problem is something that you will likely remember for the rest of your life. Very soon, you will learn an easy trick for computing the derivatives of polynomials, but for the time being, we will use definition 3 to compute this derivative. Applying definition 3, one can see that since $f(x) = x^2$, it must follow that $f(x+h) = (x+h)^2$. Plugging these into the formula for the derivative, we get

$$f'(x) = \lim_{h\to 0} \frac{f(x+h) - f(x)}{h} = \lim_{h\to 0} \frac{(x+h)^2 - x^2}{h}.$$

At this point, it may look like we are stuck and that this derivative is just equal to $\frac{0}{0}$ and that we have no hope of continuing...but fear not! We can do a little bit of algebra to see that the expression on the right hand side can be simplified by expanding out the $(x+h)^2$ term on the right hand side. Doing this, we get some nice simplification;

$$f'(x) = \lim_{h\to 0} \frac{(x+h)^2 - x^2}{h} = \lim_{h\to 0} \frac{x^2 + 2xh + h^2 - x^2}{h} = \lim_{h\to 0} \frac{2xh + h^2}{h}.$$

Ah ha! Looking at the right-most term, one can now see that we can factor out the common h from the numerator:

$$f'(x) = \lim_{h\to 0} \frac{2xh + h^2}{h} = \lim_{h\to 0} \frac{h(2x+h)}{h} = \lim_{h\to 0} 2x + h.$$

Plugging 0 for h in the last term of this limit, we can finally see that

$$f'(x) = \lim_{h\to 0} 2x + h = 2x.$$

What we just determined is that if $f(x) = x^2$, the derivative of $f(x)$, which is called $f'(x)$, is equal to 2x.

□

Now that we've computed this derivative, I want to make a quick aside regarding the notation for a derivative. In practice, if we are taking the derivative of a function $y = f(x)$, the symbol we use to denote the derivative can be one of the following:

$$f'(x) \qquad \frac{dy}{dx} \qquad \frac{df(x)}{dx}.$$

To see how to use this notation, take the previous example, where we computed the derivative of x^2 with respect to x, and found it to be $2x$. The way we would

apply this new notation is to say that if $f(x) = x^2$, then the derivative of $f(x)$ with respect to the variable x can be written as

$$f'(x) = 2x$$

$$\frac{dy}{dx} = 2x$$

or

$$\frac{df(x)}{dx} = 2x.$$

Each of these three ways of writing the derivative is acceptable, and each will serve its purpose at different points throughout your journey of calculus and physics.

Now that you have seen an example of using the definition of the derivative to compute the derivative of x^2, I'm going to introduce a much quicker way of computing the derivative of *any* polynomial. I am going to present three theorems to you, all with proofs. Reading the proofs of these theorems is optional, and will certainly not be necessary for your understanding of physics, but I don't like to present information without giving you sufficient justification, so if you are interested, you may feel free to follow along through the proofs.

Theorem 1. *Let $f(x) = ax^n$, where a and n are any real numbers. Then $f'(x) = nax^{n-1}$.*

Proof. We will prove this theorem by using the definition of the derivative. Fortunately for us, since we are applying this definition for all numbers a and n, our result will be a formula that we can use for any monomial. Before we plug everything into the definition of the derivative, let's write out the important terms. If $f(x) = ax^n$, it follows that $f(x+h) = a(x+h)^n$. You may recall that we can expand out $f(x+h)$ using the binomial theorem;

$$f(x+h) = a(x+h)^n = a\left[\binom{n}{0}x^n + \binom{n}{1}x^{n-1}h + \binom{n}{2}x^{n-2}h^2 + \cdots + \binom{n}{n-1}xh^{n-1} + \binom{n}{n}h^n\right],$$

where $\binom{n}{k}$ is the binomial coefficient and is defined as

$$\binom{n}{k} =_n C_k = \frac{n!}{k!(n-k)!}.$$

This means that

$$f(x+h) - f(x) = a\left[\binom{n}{1}x^{n-1}h + \binom{n}{2}x^{n-2}h^2 + \cdots + \binom{n}{n-1}xh^{n-1} + \binom{n}{n}h^n\right],$$

so

$$\lim_{h\to 0}\frac{f(x+h)-f(x)}{h} = \lim_{h\to 0}\frac{a}{h}\left[\binom{n}{1}x^{n-1}h + \binom{n}{2}x^{n-2}h^2 + \cdots + \binom{n}{n-1}xh^{n-1} + \binom{n}{n}h^n\right].$$

Pulling out and cancelling the common h gives,

$$f'(x) = \lim_{h\to 0} a\left[\binom{n}{1}x^{n-1} + \binom{n}{2}x^{n-2}h + \cdots + \binom{n}{n-1}xh^{n-2} + \binom{n}{n}h^{n-1}\right].$$

Each term in the previous expression, except for the first has an h in it, so since we are taking the limit as h goes to 0, each term cancels out except for the first, leaving us with

$$f'(x) = \lim_{h\to 0} a\binom{n}{1}x^{n-1} = anx^{n-1},$$

completing the proof. □

Remark. *If $f(x) = c$, where c is any real number constant, then f'(x)=0.*

Now that we have proved this formula, we can use it for any monomial. Let's look at a few examples:

Example 6. *Let $f(x) = x^3$. Find $f'(x)$.*

Solution: Theorem 1 told us that if $f(x) = ax^n$, then $f'(x) = nax^{n-1}$. This example is an special case of theorem 1 with $a = 1$ and $n = 3$, so plugging these values in, it is clear that $f'(x) = 3x^{3-1} = 3x^2$.

□

Example 7. *Find the derivative with respect to x of $f(x) = -4$*

Solution: The remark above tells us that for any constant, $f'(x) = 0$, thus if $f(x) = -4$, $f'(x) = 0$.

□

Example 8. *Find the derivative with respect to x of $f(x) = 3x^{100}$*

Solution: By theorem 1 $a = 3$, and $n = 100$, so $f'(x) = 300x^{99}$.

□

Theorem 1 is indeed quite general and actually does not limit us only to positive integer powers:

Example 9. *Find dy/dx for $y = \sqrt{x}$*

Solution: Recall that $\sqrt{x} = x^{1/2}$, so we can apply theorem 1 by setting a equal to 1, and $n = 1/2$. Doing this, we get

$$\frac{dy}{dx} = \frac{1}{2}x^{\frac{1}{2}-1} = \frac{1}{2}x^{-\frac{1}{2}} = \frac{1}{2x^{\frac{1}{2}}} = \frac{1}{2\sqrt{x}}.$$

□

Example 10. *Compute the derivative of $f(x) = 2/x^3$.*

Solution: Recall that we can write $f(x) = 2/x^3$ as $f(x) = 2x^{-3}$. Now it should be easy for you to apply theorem 1 to solve this problem. Notice that $n = -3$ and $a = 2$, so

$$f'(x) = -6x^{-3-1} = -6x^{-4} = \frac{-6}{x^4}.$$

□

Now that you have a good idea of how to compute derivatives of monomials, let's generalize this to polynomials with the remaining two theorems.

Theorem 2.

$$\frac{d}{dx}[f(x) + g(x)] = f'(x) + g'(x).$$

The statement above (called the sum rule) is to be read, "the derivative of $f(x)$ plus $g(x)$ is equal to the derivative of $f(x)$ plus the derivative of $g(x)$."

Proof. Though this statement may sound obvious, it actually does require proof. Fortunately, this proof falls straight out of the definition of the derivative:

$$\lim_{h\to 0} \frac{[f(x+h) + g(x+h)] - [f(x) + g(x)]}{h} = \lim_{h\to 0} \left(\frac{f(x+h) - f(x)}{h} + \frac{g(x+h) - g(x)}{h} \right)$$
$$= f'(x) + g'(x).$$

□

We are now equipped to prove the following theorem, which will allow us to compute the derivative of any polynomial.

Theorem 3. *Let $f(x)$ be a polynomial of any order, that is, $f(x) = a_1x^n + a_2x^{n-1} + \cdots + a_{n-1}x + a_n$, where $\{a_1, \ldots, a_n\}$ are all real coefficients. Then $f'(x) = na_1x^{n-1} + (n-1)a_2x^{n-2} + \cdots + a_{n-1}$.*

Proof. Using theorem 2, we can apply theorem 1 to each term of this polynomial, and the proof is complete. □

Let's see some examples:

Example 11. *Compute the derivative of $4x^3 + 3x$, with respect to x.*

Solution: Theorem 3 showed us that to compute this derivative, we can simply apply theorem 1 to each term of the function, so $f'(x) = 12x^2 + 3$.

Example 12. *Let $q(r) = r^4 + 6r^3 - 3/r$. Find the derivative of q with respect to r.*

Solution: Once again, we can evaluate this derivative term by term. The derivative of the first term will be $4r^3$, the derivative of the second term will be $18r^2$, and the derivative of the last term will be $3/r^2$ (remember when differentiating the last term, you are taking the derivative of $-3r^{-1}$), so combining these three terms together gives

$$q'(r) = 4r^3 + 18r^2 + \frac{3}{r^2}.$$

□

3.3 Exercises

Differentiate the following functions.

1. $f(t) = 2 - \frac{2}{3}t$
2. $f(t) = \frac{1}{2}t^6 - 3t^4 + t$
3. $y = x^{5/3} - x^{2/3}$
4. $f(r) = \pi r^2$
5. $f(r) = \frac{4}{3}\pi r^3$
6. $f(x) = x^4 - \sqrt{4x} + \frac{6}{x^5}$
7. $g(x) = (x + 4)(2x + 1)$
8. $y = (1 + x + x^2)(2 - x^4)$
9. (a) Graph the function $f(x) = 2x^2$ along the domain $0 \leq x \leq 4$.
   (b) Compute the derivative of $f(x)$ with respect to x.
   (c) Graph the function for $f'(x)$. along the same domain as in part (a).
   (d) Compute the **second derivative** of $f(x)$ (the second derivative can be denoted by $f''(x)$, and can be computed by taking the derivative of $f'(x)$).
   (e) Graph the function for $f''(x)$ along the same domain as in parts (a) and (c).
10. Figure 1 shows the displacement vs. time graph of an object.

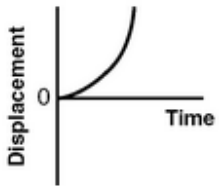


Figure 1: Figure for problem 10

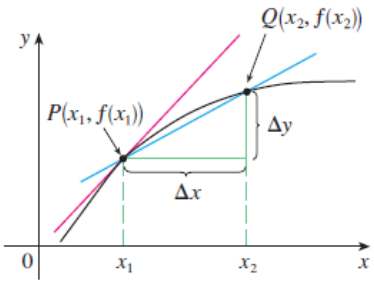
   (a) Sketch a velocity vs. time graph for this object.
   (b) Based off of what you drew in part (a), as well as your answers to problem 9; in terms of derivatives, what can be said about how velocity relates to displacement?
   (c) Sketch an acceleration vs.time graph for this object.
   (d) In terms of derivatives, what can be said about how acceleration relates to both position and velocity?

3.4 Interpreting Derivatives

Now that you know how to calculate a derivative, it's time to talk about what derivatives actually *are*, as well as how to interpret them. I previously alluded to derivatives being a rate of change, and though this was a vague description, in loose terms, that is exactly what they are—a derivative gives the slope of any function. This might not sound very significant at this point, but you will soon find that the applications of derivatives are virtually endless and are not just limited to mathematics and the physical sciences. Using derivatives, one can solve optimization problems, i.e. problems that involve finding the "best" way to do something, which is indispensible in fields such as business, finance, accounting, economics, and engineering. Unfortunately, we won't spend any time solving optimization problems in this class, but you will encounter some of these in AP Calculus, as well as in college if you choose any of these career tracks.

3.4.1 Tangent Lines and Rates of Change

No discussion about interpreting derivatives would be complete without discussing the notion of a tangent line. You all have seen tangent lines at one point or another in your mathematical careers, and we will now see exactly how tangent lines play a role into derivatives and rates of change. In the image below, let $P(x_1, f(x_1))$, and $Q(x_2, f(x_2))$ denote two points on the function drawn in black, $f(x)$.



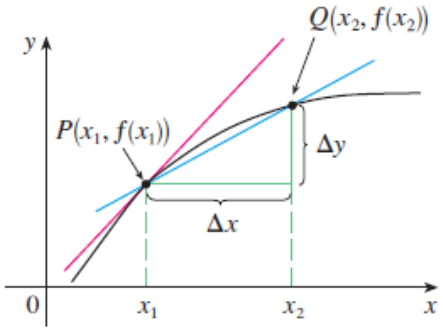
In this picture, we define the **average rate of change** of $f(x)$ to be the slope of the blue line (called the secant line) passing through points $P(x_1, f(x_1))$ and $Q(x_2, f(x_2))$. Notice that if we let $\Delta y = f(x_2) - f(x_1)$ and $\Delta x = x_2 - x_1$, it becomes clear that

$$\text{Average Rate of Change of } f(x) = \frac{\Delta y}{\Delta x} = \frac{f(x_2) - f(x_1)}{x_2 - x_1}$$

In general, the average rate of change can be defined as follows:

Definition 4. *Given a function $f(x)$, the **average rate of change** of $f(x)$ between two points P and Q is given by the slope of the secant line that connects points P and Q.*

Now let's look at the magenta line in the previous image. The image is reproduced here for your convenience.



The magenta line shows the tangent line of $f(x)$ at point P. One way to obtain this tangent line is by shrinking the interval between x_1 and x_2. Imagine we are taking the average rate of change between points P and Q, like we did before, but now we are free to move point Q along the curve of $f(x)$. If we start to move point Q closer to point P, x_2 will start getting closer and closer to x_1. If we keep moving point Q until it is directly on top of point P, P and Q will now be the same point, so we will no longer be taking the average rate of change between points P and Q, but rather, we will be finding the **instantaneous rate of change** at point P. This reasoning is motivation for the following definition:

Definition 5. *The* ***instantaneous rate of change*** *of a function $f(x)$, at some point P, is given by the slope of the tangent line of $f(x)$ at point P. In mathematical terms, we can say*

$$\textit{Instantaneous rate of change of } f(x) = \lim_{x_2 \to x_1} \frac{\Delta y}{\Delta x} = \lim_{\Delta x \to 0} \frac{\Delta y}{\Delta x}.$$

It turns out that the derivative of a function, which we had been calculating in the previous section is given by the slope of the tangent line to a curve.

Definition 6. *The derivative $f'(a)$ is the instantaneous rate of change of $y = f(x)$ with respect to x when $x = a$.*

Now that we've distinguished average rates of change from instantaneous rates of change, let's see how we can apply these to physics.

3.4.2 Velocity

Now that we have some knowledge of how to represent a derivative, we can give a much more meaningful discussion of velocity. If we let x represent position,

and t represent time, you may recall that we can define the average velocity, v_{avg} as

$$v_{avg} = \frac{\Delta x}{\Delta t} = \frac{x_2 - x_1}{t_2 - t_1} \qquad \text{(average velocity)}.$$

If your answer to exercise 10 in section 3.3 was correct, you would have found that the derivative of the displacement vs. time graph gave you velocity. It turns out that this is true for all displacement vs. time graphs. This means that if we are able to compute the derivative of a position (or displacement) vs. time function of a particle, we are able to know the instantaneous velocity, v, of this particle at any point in time.

$$v = \lim_{\Delta t \to 0} \frac{\Delta x}{\Delta t} = \frac{dx}{dt} \qquad \text{(instantaneous velocity)}.$$

Remember that dx/dt is to be read as "the derivative of the function $x(t)$ with respect to time."

Now that we have determined that velocity is given by the time derivative of position, let's see an example of exactly how to apply these principles.

Example 13. *Suppose a bird's motion is described by the position vs. time function $x(t) = 2t^3 + 4t + 1$, where x is given in meters, and t is given in seconds.*

(a) What is the average velocity of this bird between $t = 0s$ and $t = 2s$?

(b) What is the instantaneous velocity of this bird at $t = 1s$?

(c) Is the velocity of this bird constant or changing?

(d) At $t = 1$ is the bird moving in the positive x direction, or negative $x$direction?

Solution:

(a) To compute v_{avg}, we simply have to use

$$v_{avg} = \frac{\Delta x}{\Delta t}.$$

First, let's find Δx. Δx in this problem is simply the difference between our position function, $x(t)$ being evaluated at $t = 2s$, and $t = 0s$, in other words, $\Delta x = x(2) - x(0)$. Plugging in $2s$ for t, one finds that $x(2) = 2(2)^3 + 4(2) + 1 = 25$m, and plugging in $0s$ gives $x(0) = 1$m. Clearly,

$$v_{avg} = \frac{\Delta x}{\Delta t} = \frac{x(2) - x(0)}{2s - 0s} = \frac{24m}{2s} = 12\text{m/s}.$$

(b) To compute the instantaneous velocity, we have to take the derivative of $x'(t)$. Taking this derivative will give us our velocity vs. time function, $v(t)$.

$$v(t) = x'(t) = \frac{dx}{dt} = 6t^2 + 4$$

Solving for the instantaneous velocity at $t = 1$ simply requires us to evaluate $v(1) = 6(1)^2 + 4 = 10\text{m/s}$. Prior to knowing calculus, you would have likely estimated the velocity of the bird at $t = 1\text{s}$ by calculating the average velocity of the bird between $t = 0\text{s}$ and $t = 2\text{s}$, which as you can see would not be an accurate estimate, because in part (a), we found the average velocity of the bird over this time interval to be 12m/s, whereas the instantaneous velocity of the bird at $t = 1\text{s}$ is 10m/s.

(c) In part (b), we found that $v(t) = 6t^2 + 4$. Since this function still depends on t, and is not equal to a constant, it follows that in this case, v is not constant, but rather, varies quadratically with time.

(d) At $t = 1s$ we found that v was equal to $+10\text{m/s}$ which shows that the bird was moving in the positive x-direction. In order for the bird to be moving in the $-x$-direction, we would have to have v be negative at $t = 1\text{s}$.

□

3.4.3 Exercises

1. If a particle's position is given by $x = 4 - 12t + 3t^2$ (where t is in seconds, and x is in meters)
   (a) What is its velocity at $t = 1\text{s}$?
   (b) Is it moving in a positive or negative direction of x just then?
   (c) What is its speed just then?
   (d) Is the speed larger or smaller at later times?
   (e) Is there ever an instant where $v = 0$? If so, calculate the time where $v = 0$.
   (f) Is there a time after $t = 3\text{s}$ where the particle is moving in the negative direction of x?
2. The position of a particle moving along the x axis is given in centimeters by $x = 9.75 + 1.50t^3$, where t is in seconds. Calculate
   (a) Average velocity during the time interval $t = 2\text{s}$ to $t = 3\text{s}$.
   (b) The instantaneous velocity at $t = 2.0\text{s}$, $t = 2.5\text{s}$, and $t = 3\text{s}$.
   (c) Graph x versus t, and indicate your answers to the preceding parts graphically (i.e. draw tangent lines where necessary).

3.4.4 Acceleration

Thinking back to problem 10 in section 3.3, if you answered the problem correctly, you would have determined that acceleration is the derivative of velocity

vs. time and the *second* derivative of displacement vs. time. Making this formal, we can write

$$a_{avg} = \frac{\Delta v}{\Delta t} = \frac{v_2 - v_1}{t_2 - t_1} \qquad \text{(average acceleration)},$$

and

$$a = \lim_{\Delta t \to 0} \frac{\Delta v}{\Delta t} = \frac{dv}{dt} \qquad \text{(instantaneous acceleration)}.$$

When we say $a = dv/dt$, we are saying that the acceleration of an object is the derivative of the object's velocity function, with respect to time. Notice also that since $v = dx/dt$, we can write

$$a = \frac{dv}{dt} = \frac{d\frac{dx}{dt}}{dt} = \frac{d}{dt}\frac{dx}{dt} = \frac{d^2x}{dt^2}.$$

The statement above tells us that a is the second derivative of position (or displacement) with respect to time. Whenever you talk about the second derivative of a function $y = f(x)$, the notation you will use is either

$$f''(x) \qquad \text{or} \qquad \frac{d^2y}{dx^2},$$

so in the context of a position function $x(t)$, we would denote the second derivative of $x(t)$ by

$$x''(t) \qquad \text{or} \qquad \frac{d^2x}{dt^2}.$$

Example 14. *A particle's position is given by*

$$x(t) = 4 - 12t + 2t^3$$

.

(a) Find the particle's velocity function $v(t)$ and acceleration function $a(t)$.

(b) Is there ever a time where $v = 0$?

(c) Is there ever a time where $a = 0$?

Solution:

(a) We know that $v(t) = dx/dt$, so taking the derivative of $x(t)$ gives

$$v(t) = -12 + 6t^2.$$

Now, $a(t) = dv/dt = d^2x/dt^2$. Since we already computed $v(t)$, it is most convenient to find $a(t)$ by taking the time derivative of $v(t)$ (in fact you will always calculate second derivatives by taking a single derivative twice!). Doing this gives

$$a(t) = 12t$$

.

(b) To figure out whether there is a time where $v = 0$, set $v(t)$ equal to zero. Doing this gives

$$-12 + 6t^2 = 0.$$

Solving for t gives $t = \pm\sqrt{2}$, but since time can never be negative, the only solution we keep is $t = +\sqrt{2}$, so the particle's velocity is 0 at $t = \sqrt{2}$.

(c) To find out whether a is ever equal to zero, all you have to do is set $a(t)$ equal to zero and solve for t:

$$12t = 0,$$

clearly this is only true when $t = 0$, so $a = 0$ only when $t = 0$.

□

3.4.5 Exercises

1. A proton moves along the x-axis according to the equation

$$x = 50t + 10t^2,$$

where x is in meters, and t is in seconds.

(a) Calculate the average velocity of the proton during the first 3.0s of its motion.

(b) Calculate the instantaneous velocity of the proton at $t = 3.0$s.

(c) Calculate the instantaneous acceleration of the proton at $t = 3.0$s.

(d) Graph x vs. t on the domain $0 \leq t \leq 3$s. Indicate your answer to part (b) on this graph.

(e) Graph v vs. t on this same domain, and indicate your answer to part (c) on this graph.

2. An electron has a position function, in meters, of

$$x(t) = t^2 - 2t.$$

How far is this electron from the origin when it momentarily stops? (*hint*: Find when the velocity of this electron is zero, and plug that time back into your position vs. time function.)

4 Vectors

4.1 Introduction and Refresher

Recall that a vector is a quantity that has both magnitude and direction. In this level of physics, we choose to represent vectors as an arrow, with the magnitude of the vector being represented by the length of the arrow, and the direction being represented by the arrowhead. In this text, I will denote a vector by

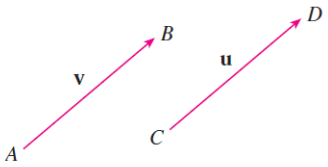


Figure 2: Equivalent Vectors

printing the variable symbol in boldface (**v**), however, when you write vector quantities, you will put an arrow above the variable symbol ($\overrightarrow{v}$) to denote a vector.

If you recall, you can add vectors by arranging them tip to tail. Another equivalent way of adding vectors is by using the **parallelogram rule** (described in the figure below).



Figure 3: Two ways of finding a resultant vector. (Left) Combining **u** and **v** by arranging the vectors tip to tail. (Right) Finding the resultant vector using the parallelogram method. You may use this method when your two vectors are arranged tail to tail. To find the resultant, simple draw a parallelogram as shown, and your resultant will be the diagonal of the parallelogram.

To find the difference between two vectors, you may simply use the identity

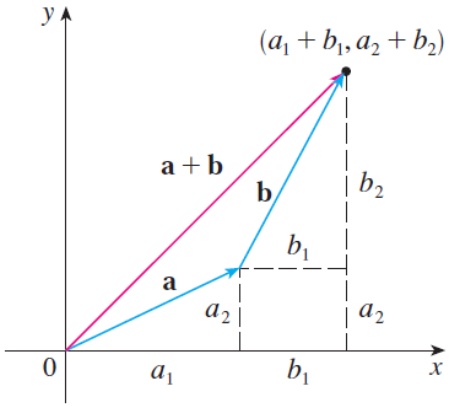
$$\mathbf{u} - \mathbf{v} = \mathbf{u} + -(\mathbf{v}),$$

thus subtracting these two vectors simply amounts to flipping the direction of vector **v**, and then adding them up either by arranging them tip to tail, or by using the parallelogram method.

Sometimes it is also useful to split vectors up into **components**. Consider the figure on the next page:

This picture shows two vectors, **a**, and **b** that are split up into their individual components, a_1 and a_2 for **a**, and b_1 and b_2 for **b**. Looking at this picture, if we assume that vector **a** makes an angle of θ_1 to the horizontal, and vector **b** makes an angle of θ_2 to the horizontal, it takes only a quick calculation to show that

$$a_1 = |\mathbf{a}| \cos\theta_1$$
$$a_2 = |\mathbf{a}| \sin\theta_1$$



$$b_1 = |\mathbf{b}| \cos \theta_2$$
$$b_2 = |\mathbf{b}| \sin \theta_2,$$

where $|\mathbf{a}|$ and $|\mathbf{b}|$ denote the **magnitude** (or length) of vectors $\mathbf{a}$ and $\mathbf{b}$, respectively. For any vector, $\mathbf{v}$, its magnitude can be calculated as

$$|\mathbf{v}| = \sqrt{v_1^2 + v_2^2},$$

where v_1 and v_2 are the x and y components of $\mathbf{v}$, respectively. A common notation for expressing a vector in terms of its components is by writing something like $\mathbf{u} =< u_1, u_2 >$. For example, if we were to say that $\mathbf{a} =< 3, 4 >$, this would mean that $\mathbf{a}$ is a vector with a magnitude of 3 units in the x direction, and a magnitude of 4 units in the y direction.

Example 15. *Calculate the magnitude of* $\mathbf{r} =< 6, -2 >$*, and determine the angle* $\mathbf{r}$ *makes with the horizontal.*

Solution: We can solve for the magnitude of $\mathbf{r}$ by taking

$$|\mathbf{r}| = \sqrt{r_1^2 + r_2^2} = \sqrt{6^2 + (-2)^2} = \sqrt{36 + 4} = \sqrt{40}$$

so the magnitude of $\mathbf{r}$ is $\sqrt{40} =\approx 6.32$ units.

To determine the angle, θ, that $\mathbf{r}$ makes to the horizontal, you could draw a right triangle with the side adjacent to θ being 6 units to the right, the side opposite to θ being 2 units down, and the hypotenuse connecting the origin to the end of the opposite side having a length of $\sqrt{40}$ units. If you do this, you will see that

$$\theta = \tan^{-1}\left(\frac{-2}{6}\right) = -18.4^{\circ},$$

or 18.4 degrees below the horizontal.

□

4.2 Unit Vectors and Vector Addition/Subtraction in Component Form

Another convenient and widely used notation to describe vectors implements the notion of a **unit vector**. A unit vector is simply a vector with a magnitude of 1. In two dimensions, the **unit basis vectors** are defined as follows:

$$\hat{\mathbf{i}} =< 1, 0 >$$

$$\hat{\mathbf{j}} =< 0, 1 >$$

Looking at these two expressions, one can see that $\hat{\mathbf{i}}$ is a vector with an x-component of 1, and a y-component of 0, and $\hat{\mathbf{j}}$ is a vector with an x-component of 0, and a y-component of 1.

We can also express vectors in a three dimensional coordinate system. In three dimensions, the unit basis vectors are

$$\hat{\mathbf{i}} =< 1, 0, 0 >$$

$$\hat{\mathbf{j}} =< 0, 1, 0 >$$

and

$$\hat{\mathbf{k}} =< 0, 0, 1 >$$

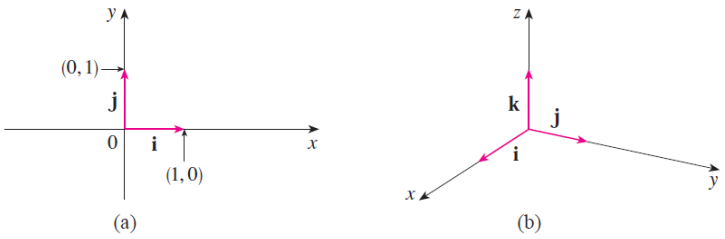


Figure 4: (a) Unit basis vectors in two dimensions. (b) Unit basis vectors in three dimensions.

Using unit vector notation, it should be clear that if $\mathbf{v} =< 1, 2, 3 >$, $\mathbf{v}$ is also equal to $1\hat{\mathbf{i}} + 2\hat{\mathbf{j}} + 3\hat{\mathbf{k}}$. These notations say the same thing and can both be used interchangably.

At this point you might be thinking, "why is Mr. Schueler introducing all of this crazy notation?!" To answer this question, it is because this notation allows us to easily manipulate vectors. For instance, if we want to add (or subtract) vectors, all we have to do is add (or subtract) each individual component. The following example is meant to help illustrate this.

Example 16. *Let* $\mathbf{a} =< 2, 1, 3 >$ *and* $\mathbf{b} =< -3, 5, 2 >$. *Find (a)* $\mathbf{a} + \mathbf{b}$*, (b)* $\mathbf{a} - \mathbf{b}$*, and (c)* $3\mathbf{a} - 2\mathbf{b}$.

Solution:

(a) Add up each component individually, so;

$$\mathbf{a}+\mathbf{b} =< 2,1,3 > + < -3,5,2 >=< (2+-(3)),(1+5),(3+2) >=< -1,6,5 >$$

(b) Subtract each component individually, so;

$$\mathbf{a}-\mathbf{b} =< 2,1,3 > + < -3,5,2 >=< (2--(3)),(1-5),(3-2) >=< 5,-4,1 >$$

(c)

$$3\mathbf{a} - 2\mathbf{b} = 3 < 2,1,3 > -2 < -3,5,2 >$$

$$=< 6,3,9 > - < -6,10,4 >=< 6 - -(6), 3 - 10, 9 - 4 >=< 12,-6,5 >$$

□

Example 17. *If* $\mathbf{a} = \hat{\mathbf{i}} + 2\hat{\mathbf{j}} - 3\hat{\mathbf{k}}$ *and* $\mathbf{b} = 4\hat{\mathbf{i}} + 7\hat{\mathbf{k}}$*, express the vector* $2\mathbf{a} + 3\mathbf{b}$ *in terms of* $\hat{\mathbf{i}}$*,* $\hat{\mathbf{j}}$*, and* $\hat{\mathbf{k}}$*.*

Solution: $2\mathbf{a} = 2(\hat{\mathbf{i}} + 2\hat{\mathbf{j}} - 3\hat{\mathbf{k}}) = 2\hat{\mathbf{i}} + 4\hat{\mathbf{j}} - 6\hat{\mathbf{k}}$, and $3\mathbf{b} = 3(4\hat{\mathbf{i}} + 7\hat{\mathbf{k}}) = 12\hat{\mathbf{i}} + 21\hat{\mathbf{k}}$, so

$$2\mathbf{a} + 3\mathbf{b} = 2\hat{\mathbf{i}} + 4\hat{\mathbf{j}} - 6\hat{\mathbf{k}} + 12\hat{\mathbf{i}} + 21\hat{\mathbf{k}} = 14\hat{\mathbf{i}} + 4\hat{\mathbf{j}} + 15\hat{\mathbf{k}}.$$

□

In the case of example 17, remember, the solution $14\hat{\mathbf{i}} + 4\hat{\mathbf{j}} + 15\hat{\mathbf{k}}$ could also be written as $< 14,4,15 >$.

4.3 Exercises

1. The figure below shows three vectors. Copy the vectors in the figure and use them to draw the following vectors:

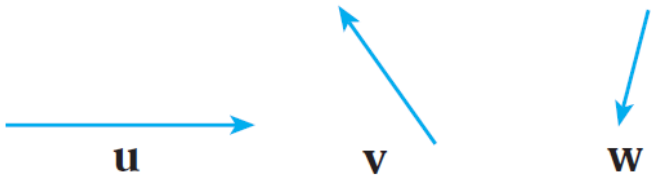


Figure 5: Figure for problem 1

   (a) $\mathbf{u} + \mathbf{v}$
   (b) $\mathbf{u} + \mathbf{w}$
   (c) $\mathbf{v} + \mathbf{w}$
   (d) $\mathbf{u} - \mathbf{v}$
   (e) $\mathbf{v} + \mathbf{u} + \mathbf{w}$
   (f) $\mathbf{u} - \mathbf{w} - \mathbf{v}$

2. Let $\mathbf{a} =< 5, -12 >$ and $\mathbf{b} =< -3, -6 >$. Find

   (a) $\mathbf{a} + \mathbf{b}$
   (b) $\mathbf{a} - \mathbf{3b}$
   (c) $|\mathbf{a}|$
   (d) $|\mathbf{a} - \mathbf{b}|$ (First calculate $\mathbf{a} - \mathbf{b}$, then find the magnitude of this result)

3. Let $\mathbf{a} = 4\hat{\mathbf{i}} + \hat{\mathbf{j}}$ and $\mathbf{b} = \hat{\mathbf{i}} - \mathbf{2}\hat{\mathbf{j}}$. Find

   (a) $\mathbf{a} + \mathbf{b}$
   (b) $\mathbf{a} - \mathbf{b}$
   (c) $|\mathbf{b}|$
   (d) $|\mathbf{a} + \mathbf{2b}|$

4. What is the angle that the vector $\hat{\mathbf{i}} + \sqrt{3}\hat{\mathbf{j}}$ makes with the positive x-axis? (*hint*: It might help to draw a picture)

4.4 Dot Products

At this point, you should have a pretty solid grasp of how to add, subtract, and find the magnitude of vectors. The next logical step is to determine how to multiply vectors. For the purposes of this class, we will be concerned with two ways of multiplying vectors: the **dot product** and the **cross product**. This section will be devoted entirely to the dot product, and the next section will be devoted entirely to the cross product. We will begin by defining the dot product.

Definition 7. *If* $\mathbf{a} =< a_1, a_2, a_3 >$ *and* $\mathbf{b} =< b_1, b_2, b_3 >$*, then the* ***dot product*** *between vectors* $\mathbf{a}$ *and* $\mathbf{b}$ *is given by:*

$$\mathbf{a} \cdot \mathbf{b} = a_1 b_1 + a_2 b_2 + a_3 b_3.$$

The expression $\mathbf{a} \cdot \mathbf{b}$ is to be read as "$\mathbf{a}$ dot $\mathbf{b}$." Notice how when computing a dot product, all you have to do is take the sum of the products of the individual components of vectors $\mathbf{a}$ and $\mathbf{b}$. This means that even though you are multiplying two vectors together, the answer is a scaler, that is, a real number. For this reason, the dot product is also sometimes called the scalar product or the inner product.

Example 18. *Let* $\mathbf{r} =< 12, -2, 6 >$ *and* $\mathbf{s} =< -0.5, 7, 2 >$*. Find* $\mathbf{r} \cdot \mathbf{s}$*.*

Solution: Calculating this dot product simply amounts to multiplying the x, y, and z components of $\mathbf{a}$ with the x, y, and ,z components of $\mathbf{b}$, respectively together, and then adding those three numbers together:

$$\mathbf{r}\cdot\mathbf{s} =< 12, -2, 6 > \cdot < -0.5, 7, 2 >= (12)(-0.5)+(-2)(7)+(6)(2) = -6+-(14)+12 = -8.$$

Example 19. *Let* $\boldsymbol{\alpha} = \hat{\mathbf{i}} - \hat{\mathbf{k}}$ *and let* $\boldsymbol{\beta} = 4\hat{\mathbf{i}} - 2\hat{\mathbf{j}} + 3\hat{\mathbf{k}}$*. Find* $\boldsymbol{\alpha} \cdot \boldsymbol{\beta}$*.*

Solution: The one tricky part about using basis vector notation, is that it may not be clear right off the bat that the $\hat{\mathbf{j}}$ component of vector $\boldsymbol{\alpha}$ is zero. Because of this, it is easier to see the solution by writing

$$\boldsymbol{\alpha} \cdot \boldsymbol{\beta} = (\hat{\mathbf{i}} + 0\hat{\mathbf{j}} - \hat{\mathbf{k}}) \cdot (4\hat{\mathbf{i}} - 2\hat{\mathbf{j}} + 3\hat{\mathbf{k}}) = 4 + 0 - 3 = 1.$$

□

The dot product can also give a geometric interpretation of the angle θ between two vectors, which is of the utmost importance in physics. This interpretation is contained in the theorem below, which can be proved using the law of cosines. I will not labor the details of the proof in this text (see your math textbook (or Mr. Schueler after school) if you are interested in the proof.)

Theorem 4. *Let* θ *be the angle between the vectors* $\mathbf{a}$ *and* $\mathbf{b}$*, then*

$$\mathbf{a} \cdot \mathbf{b} = |\mathbf{a}||\mathbf{b}| \cos\theta.$$

Proof. Look this one up in your math book or ask Mr. Schueler. □

Let's consider first, a mathematical example of applying this theorem, then a way to apply this theorem in physics.

Example 20. *If the vectors* $\mathbf{a}$ *and* $\mathbf{b}$ *have lengths of 4 and 6, and the angle between them, in radians, is* $\pi/3$*, find* $\mathbf{a} \cdot \mathbf{b}$*.*

Solution: Theorem 4 shows us that

$$\mathbf{a} \cdot \mathbf{b} = |\mathbf{a}||\mathbf{b}| \cos(\pi/3) = 4 \cdot 6 \cdot \frac{1}{2} = 12.$$

□

So why is this example important within the context of physics? Well, consider an example asking us to calculate the work done on an object. Last year, you learned that $W = Fd$, where W is work, F is force, and d is displacement. One problem with this equation is that, when you break it down, you notice that force and displacement are both vectors, so to find work, we have to multiply two vectors together. It turns out, the equation $W = Fd$ is only valid when the force and displacement vectors are parallel to each other. In general, work is actually defined as the dot product between force and displacement vectors. The next example is a restatement of example 19, this time within the context of work:

Example 21. *A 4N force acts at a 60° angle (from the horizontal) on an object and pulls it a distance of 6m. How much work is done on the object by this force?*

Solution: This example is, mathematically, identical to the previous example. Now, we know $W = \mathbf{F} \cdot \mathbf{d}$, so by theorem 4, we also have

$$W = |\mathbf{F}||\mathbf{d}| \cos\theta = (4\text{N})(6\text{m}) \cos(60^\text{o}) = 12\text{J}.$$

□

Theorem 3 can also be rearranged to solve for the angle between two vectors. This result will be stated as a corollary. In math, a corollary is a statement of interest that results from a particular theorem.

Corollary.

$$\cos\theta = \frac{\mathbf{a} \cdot \mathbf{b}}{|\mathbf{a}||\mathbf{b}|}$$

This corollary gives us some interesting insight about the dot products of vectors at various angles.

Theorem 5. *The dot product between two vectors* $\mathbf{a}$ *and* $\mathbf{b}$ *is equal to zero if and only if* $\mathbf{a}$ *and* $\mathbf{b}$ *are orthogonal (perpendicular) to one another.*

Proof. Suppose $\mathbf{a} \cdot \mathbf{b} = 0$. Then by theorem 4, we have

$$0 = |\mathbf{a}||\mathbf{b}| \cos\theta.$$

Dividing out $|\mathbf{a}||\mathbf{b}|$ gives $\cos\theta = 0$, which is true when $\theta = 90^\text{o}$, thus if $\mathbf{a}$ and $\mathbf{b} = 0$, then $\mathbf{a}$ and $\mathbf{b}$ are orthogonal.

Conversely, suppose $\mathbf{a}$ and $\mathbf{b}$ are orthogonal. Then $\cos(\theta)$ is equal to zero, thus

$$\mathbf{a} \cdot \mathbf{b} = |\mathbf{a}||\mathbf{b}| \cos(90^\text{o}) = 0,$$

which completes the proof. □

This result will prove to be particularly useful when we study Gauss' Law, as well as magnetism.

Example 22. *Find the angle between* $< 1, 2, 3 >$ *and* $< 3, -4, 6 >$.

Solution: The corollary to theorem 4 showed us that

$$\cos\theta = \frac{\mathbf{a}\cdot\mathbf{b}}{|\mathbf{a}||\mathbf{b}|}.$$

This means in order to find θ, we first need to evaluate $|\mathbf{a}|$, $|\mathbf{b}|$ and $\mathbf{a}\cdot\mathbf{b}$:

$$|\mathbf{a}| = \sqrt{1^2 + 2^2 + 3^2} = \sqrt{14}$$

$$|\mathbf{b}| = \sqrt{3^2 + (-4)^2 + 5^2} = \sqrt{61}$$

and

$$\mathbf{a}\cdot\mathbf{b} = (1)(3) + (2)(-4) + (3)(6) = 13$$

thus

$$\cos\theta = \frac{\mathbf{a}\cdot\mathbf{b}}{|\mathbf{a}||\mathbf{b}|} = \frac{13}{(\sqrt{14})(\sqrt{61})}$$

so

$$\theta = \cos^{-1}\left(\frac{13}{(\sqrt{14})(\sqrt{61})}\right) = 63.6^{\text{o}}$$

□

4.5 Exercises

1. For the following vectors, find $\mathbf{a}\cdot\mathbf{b}$:

   (a) $\mathbf{a} =< 2, 3 >$, $\mathbf{b} =< -0.7, 1.2 >$
   (b) $\mathbf{a} =< 5, 3, -4 >$, $\mathbf{b} =< 9, 2, 7 >$
   (c) $\mathbf{a} = 2\hat{\mathbf{i}} + \hat{\mathbf{j}}$, $\mathbf{b} = \hat{\mathbf{i}} - \hat{\mathbf{j}} + \hat{\mathbf{k}}$
   (d) $\mathbf{a} = 3\hat{\mathbf{i}} + 2\hat{\mathbf{j}} - \hat{\mathbf{k}}$, $\mathbf{b} = 4\hat{\mathbf{i}} + 5\hat{\mathbf{k}}$
   (e) $|\mathbf{a}| = 6$, $|\mathbf{b}| = 5$, the angle between $\mathbf{a}$ and $\mathbf{b}$ is 120°
   (f) $|\mathbf{a}| = 3$, $|\mathbf{b}| = \sqrt{6}$, the angle between $\mathbf{a}$ and $\mathbf{b}$ is 45°

2. Find the angle between the vectors. Express your answer to the nearest degree.

   (a) $\mathbf{a} =< 1, 4 >$, $\mathbf{b} =< -7, 9 >$
   (b) $\mathbf{a} =< 2, 3, -1 >$, $\mathbf{b} =< -0.7, 1.2, 3 >$
   (c) $\mathbf{a} = 4\hat{\mathbf{i}} - 3\hat{\mathbf{j}} + \hat{\mathbf{k}}$, $\mathbf{b} = 2\hat{\mathbf{i}} - \hat{\mathbf{k}}$
   (d) $\mathbf{a} = \hat{\mathbf{i}} + 2\hat{\mathbf{j}} - 2\hat{\mathbf{k}}$, $\mathbf{b} = 4\hat{\mathbf{i}} - 3\hat{\mathbf{k}}$

4.6 Cross Products

The first product of vectors that we learned about, the dot product, is an operation where you input two vectors and pop out an answer that is a scalar. By contrast, the **cross product** of two vectors results in an answer that is also a vector. Because of this, cross products are sometimes called **vector products**.

Definition 8. *Let* $\mathbf{a} =< a_1, a_2, a_3 >$ *and* $\mathbf{b} =< b_1, b_2, b_3 >$*. Then, the cross product of* $\mathbf{a}$ *and* $\mathbf{b}$ *is the vector*

$$\mathbf{a} \times \mathbf{b} =< a_2b_3 - a_3b_2, a_3b_1 - a_1b_3, a_1b_2 - a_2b_1 > .$$

The term $\mathbf{a} \times \mathbf{b}$ is to be read as "$\mathbf{a}$ cross $\mathbf{b}$." Unlike the dot product which can be defined in any number of dimensions, the cross product is only defined in 3-dimensions (actually, oddly enough, the cross product is also defined in 7-dimensions, but we will not talk about that in this class).

Using definition 8 is a perfectly legitimate way to solve a cross product, however, there are ways to compute cross products that are quite a bit less cumbersome on the memory than applying definition 8. The way I (and many others) learned to compute cross products is by using the notion of a determinant.

4.6.1 Matrix Determinants

The determinant of a (2×2) matrix is defined as follows

$$\begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc.$$

For example,

$$\begin{vmatrix} 9 & 12 \\ -2 & 1 \end{vmatrix} = (9)(1) - (12)(-2) = 9 - (-24) = 9 + 24 = 33.$$

This notion of a determinant can also be extended to (3×3) matrices, which will prove useful for calculating cross products. Determinants for (3×3) (and higher order) matrices can be computed by using what is known as a cofactor expansion. The cofactor expansion of a (3×3) splits the matrix up into three (2×2) matrices, which we can use to compute the determinant:

$$\begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix} = a_1 \begin{vmatrix} b_2 & b_3 \\ c_2 & c_3 \end{vmatrix} - a_2 \begin{vmatrix} b_1 & b_3 \\ c_1 & c_3 \end{vmatrix} + a_3 \begin{vmatrix} b_1 & b_2 \\ c_1 & c_2 \end{vmatrix}.$$

Notice how when computing this determinant, we start with a_1, then multiply a_1 by the determinant obtained by deleting the row and column in which a_1 appears. We then repeat the same process for a_2 and a_3. Finally we take the a_1 term, *subtract* from it the a_2 term, then add the a_3 term onto this. For example

$$\begin{vmatrix} 1 & -1 & 2 \\ 3 & 2 & 1 \\ -4 & 5 & 2 \end{vmatrix} = 1 \begin{vmatrix} 2 & 1 \\ 5 & 2 \end{vmatrix} - (-1) \begin{vmatrix} 3 & 1 \\ -4 & 2 \end{vmatrix} + 2 \begin{vmatrix} 3 & 2 \\ -4 & 5 \end{vmatrix}$$

$$= 1(4-5) - (-1)(6-(-4)) + 2(15-(-8)) = -1 + 10 + 46 = 55.$$

4.6.2 Computing Cross Products with determinants

Equipped with the notion of the determinant, we can alternatively define the cross product between two vectors $\mathbf{a}$ and $\mathbf{b}$ as

$$\mathbf{a} \times \mathbf{b} = \begin{vmatrix} \hat{\mathbf{i}} & \hat{\mathbf{j}} & \hat{\mathbf{k}} \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix} = \hat{\mathbf{i}} \begin{vmatrix} a_2 & a_3 \\ b_2 & b_3 \end{vmatrix} - \hat{\mathbf{j}} \begin{vmatrix} a_1 & a_3 \\ b_1 & b_3 \end{vmatrix} + \hat{\mathbf{k}} \begin{vmatrix} a_1 & a_2 \\ b_1 & b_2 \end{vmatrix}.$$

Example 23. *If* $\mathbf{a} =< 2, 1, 4 >$ *and* $\mathbf{b} =< 1, 1, 2 >$*, compute* $\mathbf{a} \times \mathbf{b}$*.*

Solution: Writing out the cross product as a determinant

$$\mathbf{a} \times \mathbf{b} = \begin{vmatrix} \hat{\mathbf{i}} & \hat{\mathbf{j}} & \hat{\mathbf{k}} \\ 2 & 1 & 4 \\ 1 & 1 & 2 \end{vmatrix} = \hat{\mathbf{i}} \begin{vmatrix} 1 & 4 \\ 1 & 2 \end{vmatrix} - \hat{\mathbf{j}} \begin{vmatrix} 2 & 4 \\ 1 & 2 \end{vmatrix} + \hat{\mathbf{k}} \begin{vmatrix} 2 & 1 \\ 1 & 1 \end{vmatrix}$$

$$= (2-4)\hat{\mathbf{i}} - (4-4)\hat{\mathbf{j}} + (2-1)\hat{\mathbf{k}} = -2\hat{\mathbf{i}} + \hat{\mathbf{k}} =< -2, 0, 1 > .$$

□

Geometrically, the cross product is defined so that $\mathbf{a} \times \mathbf{b}$ is mutally orthogonal to $\mathbf{a}$ and $\mathbf{b}$. What this means is that regardless of what angle $\mathbf{a}$ makes with $\mathbf{b}$, $\mathbf{a} \times \mathbf{b}$ will always be at a 90° angle to both $\mathbf{a}$ and $\mathbf{b}$. Figure 6 describes how to determine the direction of $\mathbf{a} \times \mathbf{b}$ using the **right hand rule**. Now that the geometry of the cross product has been briefly introduced, let's prove some theorems about cross products.

Theorem 6. *The vector* $\mathbf{a} \times \mathbf{b}$ *is perpendicular to both* $\mathbf{a}$ *and* $\mathbf{b}$*.*

Proof. Recall from the previous section that the dot product between any two vectors is 0 if and only if the two vectors are perpendicular. Based off of this information, all we have to show for our theorem to be true is that $(\mathbf{a} \times \mathbf{b}) \cdot \mathbf{a} = 0$ and $(\mathbf{a} \times \mathbf{b}) \cdot \mathbf{b} = 0$. Carrying out the calculation gives:

$$(\mathbf{a} \times \mathbf{b}) \cdot \mathbf{a} = a_1 \begin{vmatrix} a_2 & a_3 \\ b_2 & b_3 \end{vmatrix} - a_2 \begin{vmatrix} a_1 & a_3 \\ b_1 & b_3 \end{vmatrix} + a_3 \begin{vmatrix} a_1 & a_2 \\ b_1 & b_2 \end{vmatrix} = 0.$$

A similar comparison can be done with $(\mathbf{a} \times \mathbf{b}) \cdot \mathbf{b} = 0$. Therefore $\mathbf{a} \times \mathbf{b}$ is perpendicular to both $\mathbf{a}$ and $\mathbf{b}$. □

The following is another useful theorem that is quite similar in nature to a theorem we discussed about the dot product. The proof of this theorem is beyond the scope of this class, so take this statement as a given.

Theorem 7. *If* θ *is the angle between* $\mathbf{a}$ *and* $\mathbf{b}$*, then*

$$|\mathbf{a} \times \mathbf{b}| = |\mathbf{a}||\mathbf{b}| \sin\theta.$$

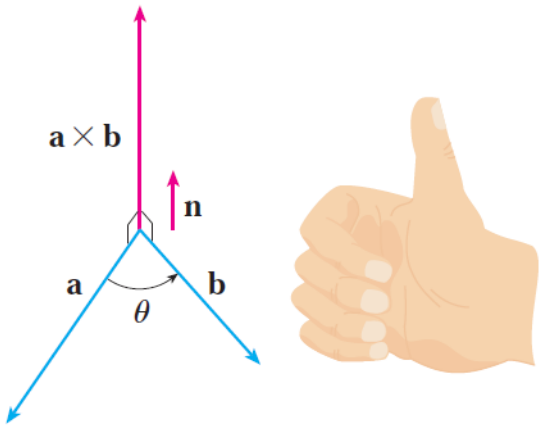


Figure 6: The geometric interpretation of a cross product. $\mathbf{a} \times \mathbf{b}$ is always perpendicular to the plane of $\mathbf{a}$ and $\mathbf{b}$. Given any vectors $\mathbf{a}$ and $\mathbf{b}$, you can determine the direction of $\mathbf{a} \times \mathbf{b}$ by using the **right hand rule**. The right hand rule works by having all of your fingers (except your thumb) pointed in the direction of vector $\mathbf{a}$. If you stick out your thumb, and curl your fingers from vector $\mathbf{a}$ to vector $\mathbf{b}$, your thumb will be pointing in the direction of $\mathbf{a} \times \mathbf{b}$. Alternatively, you could apply the right hand rule by sticking your thumb out, pointing your index finger in the direction of $\mathbf{a}$, and pointing your middle finger in the direction of $\mathbf{b}$. If this is done, your thumb will be pointing in the direction of $\mathbf{a} \times \mathbf{b}$.

We can summarize all of these findings with the following definition

Definition 9. *$\mathbf{a} \times \mathbf{b}$ is the vector that is perpendicular to both $\mathbf{a}$ and $\mathbf{b}$, whose orientation is determined by the right-hand rule (read the caption on figure 6), and whose length is $|\mathbf{a}||\mathbf{b}| \sin\theta$.*

Corollary. *Two nonzero vectors $\mathbf{a}$ and $\mathbf{b}$ are parallel if and only if*

$$\mathbf{a} \times \mathbf{b} = \mathbf{0}.$$

Proof. Two nonzero vectors are defined to be parallel if and only if $\theta = 0$ or $\theta = \pi$. In both of these cases, $\sin\theta = 0$, thus by theorem 7, $|\mathbf{a} \times \mathbf{b}| = 0$, so it follows that $\mathbf{a} \times \mathbf{b} = \mathbf{0}$. □

Up to this point, this is really all that needs to be said about cross products. You will encounter cross products in physics when we study rotational mechanics and dynamics. We will see them first when discussing torque and angular momentum. Cross products are even more prevalent when discussing magnetism and they will play a crucial role in our understanding of the physics of particles moving through magnetic fields.

4.7 Exercises

1. Find the cross product $\mathbf{a} \times \mathbf{b}$ and verify that it is perpendicular to both $\mathbf{a}$ and $\mathbf{b}$.

   (a) $\mathbf{a} =< 6, 0, -2 >$, $\mathbf{b} =< 0, 8, 0 >$

   (b) $\mathbf{a} =< 1, 1, -1 >$, $\mathbf{b} =< 2, 4, 6 >$

   (c) $\mathbf{a} = \hat{\mathbf{i}} - \hat{\mathbf{j}} - \hat{\mathbf{k}}$, $\mathbf{b} = 0.5\hat{\mathbf{i}} + \hat{\mathbf{j}} + 0.5\hat{\mathbf{k}}$

   (d) **Challenge**: $\mathbf{a} = t\hat{\mathbf{i}} + \cos t\hat{\mathbf{j}} + \sin t\hat{\mathbf{k}}$, $\mathbf{b} = \hat{\mathbf{i}} - \sin t\hat{\mathbf{j}} + \cos t\hat{\mathbf{k}}$

   For problems 2 and 3, find $|\mathbf{u} \times \mathbf{v}|$ and determine (using the right hand rule (watching a khan academy video might help with this)) whether $\mathbf{u} \times \mathbf{v}$ is directed into the plane of the page, or out of the page.

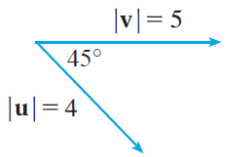


Figure 7: Figure for problem 2

2.

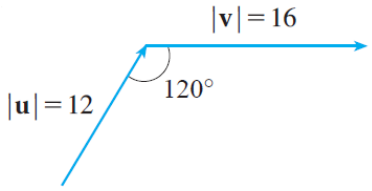


Figure 8: Figure for problem 3

3.

5 Integration

5.1 Introduction

In our discussion of differential calculus we placed a strong emphasis on idea of a rate of change. What we are going to be moving into now is the remaining branch of calculus, known as **integral calculus**. Like differential calculus, there is an idea that forms the basis of integral calculus. Rather than "breaking down" to find rates of change like we did in differential calculus, we will now be "building up" to find areas. The roots of integral calculus stem independently from Newton and Leibniz in the late 17th century, where integration was used as a tool to come up with Newton's Universal Law of Gravitation (remember that one?!). It turns out that integration and differentiation (the process of taking a derivative) are intimately related, and in some instances, an integral can be thought as an "antiderivative." Though fascinating, I will not belabor the specific theory behind integration and the integral, as it can get overly pedantic, so instead, I will show you what integration does, how to compute integrals, and why integrals are important in the applied sciences.

5.2 The Definite Integral and the Fundamental Theorem of Calculus

We will begin our discussion of integration by discussing how it can be applied to area problems. To begin, consider the following example.

Example 24. *Calculate the area under the graph of* $y = x$ *along the domain* $0 \leq x \leq 5$.

Solution: The figure below shows a graph of $y = x$ from $x = 0$ to $x = 5$. Clearly, the area under this graph is bounded below by the x-axis, so this

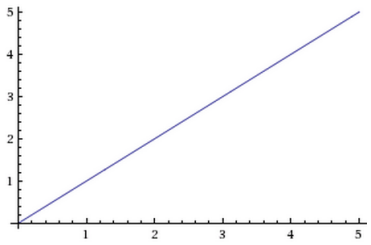


Figure 9: Graph of $y = x$ from $x = 0$ to $x = 5$

problem just amounts to us calculating the area of a triangle with a base of 5 units and a height of 5 units. If you recall from geometry, the formula for the area of a triangle is just $\frac{1}{2}bh$, where b is the base of the triangle, and h is the height of the triangle, so in this problem, the area under the curve A is given

by

$$A = \frac{1}{2}bh = \frac{1}{2}(5)(5) = 12.5.$$

□

This previous example should have been pretty straightforward. What if we wanted to find the area under a curved graph, such as the graph of $y = x^2$, for instance? In this case, the problem is not so simple and will, ultimately, require the use of integral calculus.

Without calculus, it is possible to approximate the area under curved surfaces. One way to do this would be to subdivide intervals below a curve into rectangular partitions. The figure below shows this: Figure 10 gives a decent

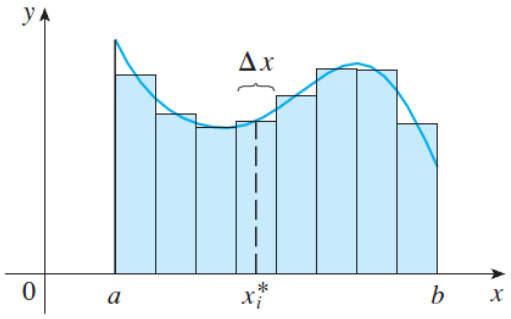


Figure 10: The area under a curve along the closed interval $[a, b]$ approximated by using 8 rectangles, with the midpoint of the top of each rectangle touching the curve.

approximation of the area under this curve, but it is still not exact. Looking at figure 8, you might be thinking, what if we approximated the with more than 8 rectangles, wouldn't that give us a more accurate area? If this is what you were thinking, you would be correct, the more rectangles we use, the better approximation we get! It turns out, that as we increase the number of rectangles

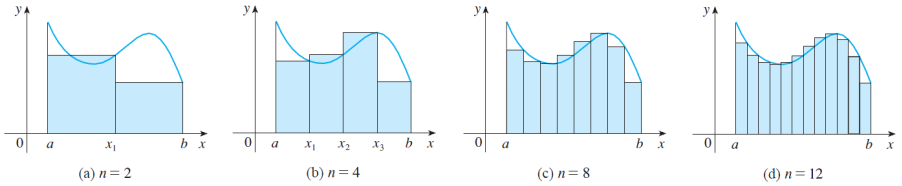


Figure 11: As we increase the number of equally spaced rectangles between a and b, we get a more accurate estimate of the area under the curve

to infinity, we will end up getting the exact area under the curve. This idea, of

placing an infinite number of infinitely thin rectangles under a curve to find the area under it, forms the basis of integration and the definite integral.

Definition 10. *Suppose we want to find the area under the curve $y = f(x)$, bounded below by the x-axis, along the closed interval $[a, b]$. The definite integral, written as*

$$\int_a^b f(x)dx,$$

gives us this area.

The symbol

$$\int_a^b f(x)dx$$

is to be read as "the integral from a to b of $f(x)$ with respect to x." The dx tells us the variable that we are integrating the function $f(x)$ over, is x. The variable a is called the **lower bound of integration**, and the variable b is called the **upper bound of integration**. Now that we have discussed the integral as

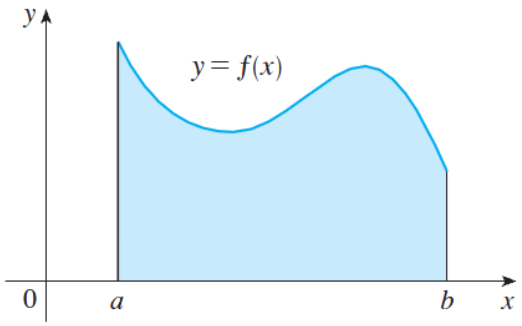


Figure 12: $\int_a^b f(x)dx$ gives the exact area under $f(x)$ from a to b

a tool for finding areas, let's see how we can compute integrals. The theorem below, known as the **fundamental theorem of calculus**, gives us an easy and effective way of computing integrals.

Theorem 8. *Suppose f is a continuous function on $[a, b]$.*

1. *If $g(x) = \int_a^x f(t)dt$, then $g'(x) = f(x)$.*
2. *$\int_a^b f(x)dx = F(b) - F(a)$, where F is any antiderivative of f, that is, $F' = f$*

Proof. See your calculus textbook. □

Theorem 8 tells us that the definite integral can function as an antiderivative. The fact that the integral serves as an antiderivative, allows us to, in a similar fashion as with the derivative, come up with a formula to solve for the integral of any polynomial.

Theorem 9. *Let $f(x) = kx^n$, then*

$$\int_a^b f(x)dx = F(b) - F(a),$$

where $F(x)$ is called the ***integrated function****, and is given by*

$$F(x) = \frac{k}{n+1}x^{n+1}.$$

Now we are equipped to use the definite integral to solve the problem from example 23.

Example 25. *Calculate the area under the graph of $y = x$ along the domain $0 \leq x \leq 5$.*

Solution: We will now solve this problem using the definite integral. Since the function we are finding the area under is $y = x$, we will set $f(x) = x$. The interval we are integrating over is $[0, 5]$, so when setting our bounds for integration, we will have $a = 0$ and $b = 5$, thus our integral will look like

$$\int_a^b f(x)dx = \int_0^5 xdx.$$

Looking back at theorem 9, we see that with $f(x) = x$, $k = 1$ and $n = 1$, so plugging these into $F(x)$ from theorem 9, we get our integrated function to be

$$F(x) = \frac{k}{n+1}x^{n+1} = \frac{1}{1+1}x^{1+1} = \frac{1}{2}x^2.$$

Now all we need to do is plug in our bounds of integration into our integrated function. Remember, by the fundamental theorem of calculus

$$\int_0^5 xdx = F(5) - F(0),$$

so since $F(x) = \frac{1}{2}x^2$, it follows that $F(5) = \frac{1}{2}(5)^2 = 12.5$, and $F(0) = \frac{1}{2}(0)^2 = 0$, thus $F(b) - F(a) = 12.5 - 0 = 12.5$, which is consistent with our answer to example 23.

□

Now that we are equipped with the definite integral, using theorem 9, we can compute the integral, and thus, the area under the curve, of any monomial.

Example 26. *Compute the area under the curve of $f(x) = 2x^2$ along the domain of $1 \leq x \leq 4$.*

Solution: In this example, we are integrating the function $f(x) = 2x^2$ over the domain $1 \leq x \leq 4$, so our lower bound of integration, a, will be 1, and our upper bound of integration, b, will be 4, so our integral will look as follows:

$$\int_1^4 2x^2 dx.$$

Theorem 9 tells us that our integrated function $F(x)$ will be of the form:

$$F(x) = \frac{k}{n+1}x^{n+1},$$

where in this problem $k = 2$ and $n = 2$, so plugging these in will give

$$F(x) = \frac{2}{3}x^3.$$

Evaluating this integral simply amounts to taking the difference between F evaluated at the upper bound of integration, and F evaluated at the lower bound of integration, hence

$$\int_1^4 2x^2 dx = F(4) - F(1) = \frac{2}{3}(64 - 1) = 42.$$

□

Sometimes problems will simply ask you to evaluate a definite integral. The example below illustrates this:

Example 27. *Compute*

$$\int_3^{12} \sqrt{x} dx.$$

Solution: Asking us to compute this integral is the same thing as asking us to calculate the area under the graph of $\sqrt{x}$ along the interval $[3, 12]$ (it actually doesn't matter if this interval is open or closed, this means that you could integrate over the domain $[3, 12)$, $(3, 12]$, or $(3, 12)$ and still get the same answer. In fact, for each of these intervals, you would plug in your bounds of integration just the same!). This problem might stump you at first glance, but it turns out theorem 9 is enough to allow us to compute this integral because $\sqrt{x} = x^{1/2}$, so we can say

$$\int_3^{12} \sqrt{x} dx = \int_3^{12} x^{1/2} dx = F(12) - F(3),$$

where our integrated function $F(x)$ is obtained by plugging in 1 for k, and $1/2$ for n into the general form given in theorem 9:

$$F(x) = \frac{1}{\frac{1}{2}+1}x^{1/2+1} = \frac{1}{\frac{3}{2}}x^{3/2} = \frac{2}{3}x^{3/2}.$$

This means

$$\int_3^{12} \sqrt{x}dx = F(12) - F(3) = \frac{2}{3}[(12)^{3/2} - (3)^{3/2}] \approx \frac{2}{3}(36.373) \approx 24.25.$$

□

Listed below is another useful theorem that will extend the class of integrals you can solve

Theorem 10.

$$\int_a^b (f(x) + g(x))dx = \int_a^b f(x)dx + \int_a^b g(x)dx.$$

Theorem 10, in words tells us that if we have any function with more than one term in it, we can evaulate the integral of each term separately. The example below will illustrate this.

Example 28. *Compute*

$$\int_{-2}^{3} (12x^2 + 2x - 6)dx.$$

Solution: Theorem 10 tells us that solving this problem amounts to computing

$$\int_{-2}^{3} 12x^2dx + \int_{-2}^{3} 2xdx - \int_{-2}^{3} 6dx.$$

A nice way to organize these functions is to compute the integrated function $F(x)$ for each function we are integrating, separately. The table below shows the integrated function $F(x)$ for each function $f(x)$: By looking at table 3, it

$f(x)$	$F(x)$
$12x^2$	$4x^3$
$2x$	x^2
6	$6x$

Table 3: The left column shows the three functions we are integrating, and the right column shows the integrated function.

should be clear that

$$\int_{-2}^{3} (12x^2 + 2x - 6)dx = F(3) - F(-2)$$

where

$$F(x) = 4x^3 + x^2 - 6x,$$

so $F(3) = 4(3)^3 + (3)^2 - 6(3) = 99$ and $F(-2) = 4(-2)^3 + (-2)^2 - 6(-2) = -16$, therefore

$$\int_{-2}^{3} (12x^2 + 2x - 6)dx = F(3) - F(-2) = 99 - (-16) = 115.$$

□

The last example we will consider here, is integrating a rational function. Like before, it turns out that theorems 9 and 10 will allow us to compute this integral with ease.

Example 29. *Compute*

$$\int_{-4}^{-1} \left(\frac{2}{x^3} + \frac{1}{x^2} \right) dx.$$

Solution: First note that $2/x^3$ can be rewritten as $2x^{-3}$, and similarly, $1/x^2$ can be rewritten as x^{-2}. Now, using theorem 10, we have

$$\int_{-4}^{-1} \left(\frac{2}{x^3} + \frac{1}{x^2} \right) dx = \int_{-4}^{-1} 2x^{-3}dx + \int_{-4}^{-1} x^{-2}dx.$$

Recall from theorem 9 that for any function $f(x) = kx^n$, the integrated function, $F(x)$ is given by

$$F(x) = \frac{k}{n+1} x^{n+1}.$$

Looking at this, it should become clear that

$$\int_{-4}^{-1} 2x^{-3}dx = F_1(-1) - F_1(-4)$$

where

$$F_1(x) = \frac{2}{-3+1} x^{-3+1} = -x^{-2} = \frac{-1}{x^2}$$

and

$$\int_{-4}^{-1} x^{-2}dx = F_2(-1) - F_2(-4)$$

where

$$F_2(x) = \frac{1}{-2+1} x^{-2+1} = -x^{-1} = \frac{-1}{x}.$$

Plugging in our numbers gives

$$\int_{-4}^{-1} 2x^{-3}dx = F_1(-1) - F_1(-4) = \frac{-1}{1} - \frac{-1}{16} = -1 + \frac{1}{16} = \frac{-15}{16}$$

and

$$\int_{-4}^{-1} x^{-2}dx = F_2(-1) - F_2(-4) = \frac{-1}{-1} - \frac{-1}{-4} = 1 - \frac{1}{4} = \frac{3}{4}.$$

Combining this together gives

$$\int_{-4}^{-1} \left(\frac{2}{x^3} + \frac{1}{x^2} \right) dx = \frac{-15}{16} + \frac{3}{4} = \frac{-3}{16}.$$

□

As you process the answer to example 28. you might be thinking "wait, if the definite integral gives me the area under the curve of a graph, how could I be getting a negative answer for this integral?!" From what I have told you so far, you should be concerned with the fact that this integral is negative. Fortunately, the reason why you are concerned is because I have not yet given you the full picture. Whenever the result of your integral is a positive number, you are computing the area under the curve, bounded below by the x-axis, whereas whenever the result of your integral is a negative number, you are computing the area of the region of a graph that lies below the x-axis, and bounded above by the x-axis. By convention, the area of a curve that lies below the x-axis and is bounded above by the x-axis is defined to be negative.

5.3 Exercises

Compute the following definite integrals.

1. $\int_{-1}^{2}(x^3 - 2x)dx$
2. $\int_{-1}^{1} x^{100}dx$
3. $\int_{0}^{1}(x+2)(x-3)dx$ (*Hint*: Expand this polynomial first, then integrate)
4. $\int_{0}^{4} x^{2/3}dx$
5. $\int_{1}^{8} x^{-2/3}dx$
6. $\int_{0}^{4}(4-x)\sqrt{x}dx$ (*Hint*: Distribute the $\sqrt{x}$ inside the parenthesis, then integrate)

5.4 The Indefinite Integral

The fundamental theorem of calculus gave us a way of expressing an integral as an antiderivative. It is important for us to develop an efficient notation to use when we express an antiderivative.

Definition 11. *When we integrate to obtain an antiderivative, we call the integral we are taking an* ***indefinite integral****, and write*

$$\int f(x)dx = F(x) \qquad \textit{where} \qquad F'(x) = f(x).$$

In words, this statement says that the integral of $f(x)$ with respect to x gives us the integrated function $F(x)$. Conversely, the derivative of our integrated function, $F'(x)$, gives us our original function $f(x)$. Another potentially important bit of vocabulary, is that when writing $\int f(x)dx$, the function that is being integrated, $f(x)$, is called the **integrand**.

The indefinite integral, to an extent, is easier to compute than the definite integral, as it requires less steps since we don't have to evaluate any bounds of integration. Let's briefly suppose that we wanted to compute

$$\int x^3 dx.$$

Theorem 9 from the previous section told us that for $f(x) = x^3$,

$$F(x) = \frac{1}{4}x^4.$$

This means that

$$\int x^3 dx = \frac{1}{4}x^4,$$

and we can check that this is consistent with definition 11, because the derivative of $\frac{1}{4}x^4$ is equal to x^3. There is another slight problem, though. Couldn't we also say that

$$\int x^3 dx = \frac{1}{4}x^4 + 1?$$

This would be consistent with definition 11, because the derivative of $\frac{1}{4}x^4 + 1$ is still equal to x^3. It turns out that $\frac{1}{4}x^4 + 1$ is an acceptable answer and in fact $\frac{1}{4}x^4 + C$, for any constant real number, C is an acceptable answer. What we say, then, is that the solution to any indefinite integral is unique up to a constant C, that is, with the exception of a constant C, there is only one solution to an integral. Tying this back to integrating x^3 with respect to x, the actual way we would write this answer is

$$\int x^3 dx = \frac{1}{4}x^4 + C,$$

where C is any real constant.

Example 30. *Integrate*

$$\int \left(\frac{1}{2}x^2 + 4\right) dx,$$

and check that your answer is a true antiderivative, that is, check that the derivative of your answer is equal to $\frac{1}{2}x^2 + 4$.

Solution: We can solve this problem by integrating term by term. For the first term, we would get

$$\int \frac{1}{2}x^2 dx = \frac{\frac{1}{2}}{2+1}x^{2+1} = \frac{1}{6}x^3,$$

and for the second term, we get

$$\int 4dx = 4x,$$

so combining these terms, we are left with

$$\int \left(\frac{1}{2}x^2 + 4\right) dx = \frac{1}{6}x^3 + 4x.$$

We are not quite done yet, as this solution is not unique. The last thing that we have to do is add C to our answer to make it unique, therefore

$$\int \left(\frac{1}{2}x^2 + 4\right) dx = \frac{1}{6}x^3 + 4x + C.$$

Now let's check that our answer $F(x) = \frac{1}{6}x^3 + 4x + C$ is indeed a true antiderivative. Differentiating term by term, it should be easy to check that $F'(x) = \frac{1}{2}x^2 + 4$, which is precisely equal to our integrand, $f(x) = \frac{1}{2}x^2 + 4$, thus definition 11 is satified, and we see that $F(x)$ is the true antiderivative of $f(x)$.

□

5.5 Exercises

Compute the following indefinite integrals.

1. $\int (x^2 + x^{-2})dx$
2. $\int (x^3 + 1.8x^2 - 2.4x)dx$
3. $\int (x + 4)(2x + 1)dx$ (*Hint*: Expand this polynomial first, then integrate)
4. $\int \left(x^2 + 1 + \frac{1}{x^2}\right) dx$
5. $\int \left(x^{3/2} + \sqrt{x} - \frac{1}{\sqrt{x}}\right) dx$
6. $\int \frac{x^3}{x^6} dx$

5.6 Integrals in Physics

Integrals come up quite frequently in physics, so let's look at a cute example of how we can use integrals to derive the kinematics equations we learned last year. Before we begin our derivation, recall that the derivative of position with respect to time gives velocity, or in other words

$$v(t) = \frac{dx(t)}{dt}.$$

Also recall that the derivative of velocity with respect to time gives acceleration,

$$a(t) = \frac{dv(t)}{dt} = \frac{d^2x(t)}{dt^2}.$$

Now that we know that integrals can give you the area under a curve, and that in addition to this, an integral can be thought of as an antiderivative, it should finally make sense why last year I told you that the area under the curve of an

acceleration vs. time graph gives you velocity, and the area under the curve of a velocity vs. time graph gives you displacement. In mathematical terms

$$v(t) \equiv \int_{t_i}^{t_f} a(t')dt',$$

and

$$x(t) \equiv \int_{t_i}^{t_f} v(t')dt'.$$

Don't worry about the complicated notation here, this is just to show you that velocity can be defined in terms of the integral of an acceleration function depending on time, and position can be defined in terms of the integral of a velocity function depending on time.

Now that all of the preliminaries are put into place let's begin our derivation of the kinematics equations.

5.6.1 Kinematics Equation 1 Derivation

The kinematics equation that we want to derive is

$$v_f = v_i + at.$$

Now that you are at a greater level of mathematical maturity, I am going to change the notation that we use for the kinematics equations to something a bit more elegant and general. If we let v_0 denote an object's velocity at time $t = 0$, and v denote the velocity at any later time, t, then we can write the first kinematics equation as

$$v = v_0 + at.$$

Derivation: Let's start with the definition of acceleration

$$a = \frac{dv}{dt},$$

multiplying both sides by dt, we can rewrite this as

$$dv = adt.$$

What we have now is called a differential equation. We can solve this differential equation by taking the indefinite integral on both sides, that is

$$\int dv = \int adt.$$

Remember, for the kinematics equations, a is just a constant, or a number, so we can pull a out of the integral sign and write

$$\int dv = a \int dt$$

or

$$v = at + C.$$

We are not quite finished yet, for we must now determine what C is. The way that we find C is by plugging in what is called an **initial condition**. For this derivation, our initial condition will be $v = v_0$ at $t = 0$, so we can start with

$$v = at + C$$

and plug in v_0 for v and 0 for t. Doing this gives

$$v_0 = a(0) + C$$

which implies that

$$v_0 = C,$$

so we can say

$$v = at + v_0,$$

which is what we were trying to derive.

5.6.2 Kinematics Equation 2 Derivation

Now let's derive

$$d = v_i t + \frac{1}{2}at^2.$$

Before we begin this derivation, like last time, I want to introduce more sophisticated notation. With this notation, let x_0 be an object's position at time $t = 0$ and let x denote the position of the object at any later time t. Then we will write the second kinematics equation as

$$x = x_0 + v_0 t + \frac{1}{2}at^2 \qquad \text{or} \qquad \Delta x = v_0 t + \frac{1}{2}at^2.$$

Derivation: We will begin this derivation with the definition of velocity,

$$v = \frac{dx}{dt}.$$

Like with the derivation of the first kinematics equation, we can rearrange this equation to obtain a differential equation;

$$dx = vdt.$$

Integrating on both sides, we get

$$\int dx = \int vdt,$$

but we know from the first kinematics equation that at constant acceleration $v = v_0 + at$, so we can plug this in for v into the equation above, giving us

$$\int dx = \int (v_0 + at)dt.$$

Expanding this out gives

$$\int dx = \int v_0 dt + a \int t dt,$$

(the a is pulled out of the integrand because a does not depend on time, so we treat it just as a number). Evaluating both integrals gives

$$x = v_0 t + a\left(\frac{1}{2}t^2\right) + C = v_0 t + \frac{1}{2}at^2 + C.$$

What remains is to plug in our intial conditions to find C. Recall that $x = x_0$ when $t = 0$, thus

$$x_0 = v_0(0) + \frac{1}{2}a(0)^2 + C$$

so $C = x_0$, leaving us with

$$x = x_0 + v_0 t + \frac{1}{2}at^2,$$

as desired.

Exercise

You goal in this exercise is to derive the final kinematics equation we talked about last year. In the notation that we are currently using, this equation would be written as

$$v^2 = v_i^2 + 2a(x - x_0).$$

(a) Rearrange $v = v_0 + at$ so that you are solving for t

(b) Substitute your result from part (a) in for t in the equation

$$x = x_0 + v_0 t + \frac{1}{2}at^2,$$

to show that

$$v^2 = v_i^2 + 2a(x - x_0).$$

6 Answers to Problems

Section 2.2 Answers

1. 0
2. 0
3. $-\frac{1}{2}$
4. 0
5. 0.125
6. 1
7. 1
8. $\frac{1}{\sqrt{2}}$
9. $\frac{2}{3}$

Section 3.3 Answers

1. $f'(t) = -\frac{2}{3}$
2. $f'(t) = 3t^5 - 12t^3 + 1$
3. $y' = \frac{5}{3}x^{2/3} - \frac{2}{3}x^{-1/3}$
4. $f'(r) = 2\pi r$. Interestingly enough, the derivative of the formula forthe area of a circle with respect to the circle's radius gives the formula for the circumference of a circle.
5. $f'(r) = 4\pi r^2$. This result is also true for relating the volume of a sphere and its surface area. This can be generalized to any number of spatial dimensions (even greater than 3!)!
6. $f'(x) = 4x^3 - \frac{1}{\sqrt{x}} - \frac{30}{x^6}$
7. $g'(x) = 4x + 9$
8. $\frac{dy}{dx} = -6x^5 - 5x^4 - 4x^3 + 4x + 2$
9. b. $f'(x) = 4x$
   d. $f''(x) = 4$
10. b. Velocity is the derivative with respect to time of displacement (or position). In other words $v(t) = x'(t)$ or $v(t) = \frac{dx(t)}{dt}$, where $v(t)$ represents a velocity function depending on time, and $x(t)$ is a position or displacement function.
    d. Acceleration is the first derivative with respect to time of velocity, and the second derivative with respect to time of position. That is, $a(t) = \frac{dv}{dt} = \frac{d^2x}{dt^2}$.

Section 3.4.3 Answers

1. a. $v(t) = x'(t)$, so take the derivative of x and plug in 1 for t. Doing this, you should get $v(1) = -6$m/s.
   b. Since $v(1)$ is negative, the particle is moving in the negative x direction.
   c. Speed is given by $|v|$, so the speed is +6m/s.
   d. At $t = 1$s, the particle is slowing down (see if you can justify this for yourself).
   e. Yes! Set your function $v(t)$ equal to zero and solve for t. You should get $v(2) = 0$.
   f. No.
2. a. $v_{\text{avg}} = 28.5$ m/s
   b. $v(2) = 18$m/s, $v(2.5) = 28.125$ m/s, and $v(3) = 40.5$ m/s.

Section 3.4.5 Answers

1. a. $v_{\text{avg}} = 80$m/s
   b. $v(3) = 110$m/s
   c. $a(3) = 30$m/s^2

2. $v(t) = 0$ at $t = 1$s (make sure you know how to figure this out!). Plugging $t = 1$s to $x(t)$ gives $x(1) = -1$m, so the particle is 1 meter from the origin.

Section 4.3 Answers

2. a. $< 2, -18 >$
   b. $< 14, 6 >$
   c. 13
   d. 10
   b. $3\hat{\mathbf{i}} + 3\hat{\mathbf{j}}$
   c. $\sqrt{5}$
   d. $3\sqrt{5}$
3. a. $5\hat{\mathbf{i}} - \hat{\mathbf{j}}$
4. 60° or $\frac{\pi}{3}$

Section 4.5 Answers

1. a. 2.2
   b. 23
   c. 1
   d. 7
   e. -15
   f. $3\sqrt{3}$
2. a. 52°
   b. 94°
   c. 52°
   d. 49°

Section 4.7 Answers

1. a. $< 16, 0, 48 >$
   b. $< 10, -8, 2 >$
   c. $0.5\hat{\mathbf{i}} - \hat{\mathbf{j}} + 1.5\hat{\mathbf{k}}$
   d. $< 1, \sin t - t\cos t, -\cos t - t\sin t >$
2. $10\sqrt{2}$, out of the page.
3. $96\sqrt{3}$, into the page.

Section 5.3 Answers

1. $\frac{3}{4}$
2. $\frac{2}{101}$
3. $-\frac{37}{6}$
4. ≈ 6.05
5. 3
6. $\frac{94}{15}$

Section 5.5 Answers

1. $\frac{x^3}{3} - x^{-1} + C$
2. $\frac{x^4}{4} + \frac{1.8x^3}{3} - 1.2x^2 + C$
3. $\frac{2x^3}{3} + \frac{9x^2}{2} + 4x + C$
4. $\frac{x^3}{3} + x - \frac{1}{x} + C$
5. $\frac{2}{5}x^{5/2} + \frac{2}{3}x^{3/2} - 2\sqrt{x} + C$
6. $-\frac{1}{2}x^{-2} + C$